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M 401 DEMINERALIZED WATER SYSTEM

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with part 4, Subsection 1.2. "Summary of Work".

The Demineralized Water Plant shall consist of:

- Three (3) new train (2 Operating, 1 standby) for Block 3 & 4 with capacity 27.5 m3/hour.
- Block 2 shall be supplied by existing demineralized water plant

The Demineralized Water System shall be furnished complete with all required equipment, motor drives, pumps, tanks, integral controls, appurtenances and accessories as hereinafter specified including but not be limited to the following:

- Three (3) new train (2 operating, 1 standby) demineralized water plant shall consisting of mixed bed vessels complete with internals and initial charges of resin.
- Two (2) x 100% Capacity Water treatment supply pump and motor.
- Three (3) Pre-filters (2 operating, 1 standby)
- The regeneration equipment consisting of the following:
 - (a) Acid and caustic bulk storage tanks
 - (b) One (1) caustic hot water tank (if required)
 - (c) One (1) acid day tank
 - (d) One (1) caustic day tank
 - (e) Two (2) x 100% acid feed metering pumps
 - (f) Two (2) x 100% caustic feed metering pumps
- Two (2) x 100% make up water transfer pumps per each **block 2, 3 & 4.**
- All required piping controls and instrumentation for the efficient and safe operation of the plant.
- All gallery work for access to instruments and equipment for inspection and maintenance.

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- All electrical equipment with equipment as defined in Clause 8.
- Shop and field painting.
- Control panels as described in Clause 7 of this Technical Specification.

Piping and valves, foundation bolts and nuts, platforms. insulation, painting, cables, and special tools and standard tools.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 “Codes and Standards” are specifically applicable to the design manufacture and testing of the equipment included in this Specification. All references to the following technical publications are to the latest issue of each together with the latest additions of each or amendments there to unless otherwise noted.

AGMA	American Gear Manufacturer Association
AWWA	American Water Works Association
AISC	American Institute of Steel Construction
AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
APHA	American Public Health Association
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for testing and Material
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
NEMA	National Electrical Manufacturers Association

SSPC	Structural Steel painting Council
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3. DESIGN AND OPERATING CONDITIONS

3.1 Design Criteria

3.1.1 Process Requirements

The demineralization process shall include the processes described below in their respective order. Mixed bed polishers shall be cation-anion mixed bed system.

At pre-starting, standby and low flow or during short shut downs, the mixed bed effluent shall be recycled to the raw water tank.

Each mixed bed polisher shall be regenerated on a routine basis as determined by operation.

Mixed bed polishers shall be regenerated with hydrochloric acid and sodium hydroxide.

3.1.2 Equipment Internals

Design Loads: The header pipes and laterals of the distributors, collectors and under-drains of each ion exchange vessel shall be designed to adequately withstand all loads expected during normal operation and all possible transients. These shall include but not be limited to the following loads:

- (a) The dead weight of water and ion exchange media acting on the projected area of the header pipes and laterals.
- (b) Thermal loads.
- (c) Hydrodynamic loads due to the flow of water and drag loads due to the ion exchange media when the maximum allowable differential pressure across the ion exchange media has been reached or exceeded.
- (d) Osmotic loads or forces on the distributors, collectors and under-drains of the depth of resin undergoing regeneration, while remaining compacted following a service cycle.
- (e) Nozzle loads transmitted by the header piping.
- (f) The following loads may be treated as static loads with suitable

dynamic load factors:

- Flow induced vibrations during startup occasional pressure surges in the system.
- Transient loads during regeneration cycles.

Stress Concentration Effect: Stress concentration effects shall be conservatively calculated at all applicable sections. The highly stressed sections between the header pipe and central distributor pipe and between the header pipe and laterals shall be adequately evaluated.

Fatigue Evaluation: Under fluctuating loads, the fatigue effect of cantilever bending stresses in the header pipes and laterals shall be considered. Adequate provision shall be made for long fatigue life.

Stress Corrosion Effect: Due to continuous chemical processes occurring during use of this equipment, stress corrosion effects shall be evaluated. All possibilities of stress corrosion cracking shall be eliminated.

3.1.3 Operation Safety

Preparation of acid and caustic regenerant solutions shall be automatic, safe and trouble-free. The Bidder shall ensure that all safety regulations have been considered in the design of any equipment intended to store, transfer or otherwise contact acid and caustic. Concentrated acid, caustic and dilution water piping shall be furnished with flow switches to automatically trip the metering pumps on loss of either chemical feed or dilution water flow. Double check valves shall be furnished in the acid, caustic and dilution water piping at the mixing tees. Double block and bleed valves shall be furnished for the concentrated acid and caustic piping. Concentrated acid and caustic piping shall also be furnished with suction strainers, metering pump outlet check and relief valves, pressure gauges, backpressure valves and pulsation dampers. Rate of flow indicating gauges shall be furnished for local mounting for the dilute chemical piping.

Dilution of both caustic and acid shall utilize demineralized water furnished from the Demineralized storage tank.

3.1.4 Plant Arrangement

The Bidder shall design the Interconnecting piping, controls instrumentation

and regeneration systems, so this plant arrangement can be made at a minimum total cost for equipment and installation.

The arrangement of the equipment, including piping and auxiliaries, shall be developed by the Bidder and reviewed by the Owner. The arrangement shall provide adequate clearance areas and safe access for operation and maintenance.

3.2 Design Data

3.2.1 Demineralized Water Equipment

- (a) Installation : Steel Structure (Shelter Type)
- (b) Type : Ion Exchange resin mixed bed with pre-filter
- (c) Number : Three (3) train (2 operating, 1 standby)
- (d) Capacity : 3 x 27.5 m³/h (new).
- (e) Quality of Treated Water :
 - Conductivity : Less than 0.1 micro-mho/cm at 25 °C.
 - Dissolved solids : Less than 0.01 ppm as CaCO₃
 - Silica : Less than 0.01 ppm
 - Total Fe : Less than 0.01 mg/l
 - Total Cu : Less than 0.005 mg/l
 - pH : 6.5 - 7.5
- (f) Life of the pre filter : 2 month or more
- (g) Quality of raw water/distillate water :
 - Total dissolved solid : Less than 10 ppm
 - Fe : Less than 0.2 ppm
 - pH : 6.5 – 7.0

Other qualities shall be based on those of the Desalination plant manufacturer's data product water quality.
- (h) Heater for caustic soda tank (if applicable).
- (i) Regenerative chemical : HCl 35%, NaOH 45%

(j)

3.2.2 Make-Up Water Transfer Pump

- (a) Type : Horizontal centrifugal
- (b) Number : Two (2) sets per each block 3 & 4, 100% capacity each
- (c) Capacity : Sufficient for water treatment equipment and transfer demineralized water to make-up water storage tank.

3.2.3 Operating Conditions

The feed to the demineralized plant shall be from the Raw Water Tanks. The Demineralized water shall be used as make up/feedwater to dual/triple pressure heat recovery steam generator and Steam Turbine Closed Cooling Water and others.

The Demineralized Water System shall be guaranteed to produce demineralized water to support operation of the power plant. Any quantity of water required for regeneration and other uses shall be added to the net demineralized water required to verify total flow requirements.

Installation and trial operation of the Demineralized Water System shall be completed before the date when Steam Turbine operate and the plant shall be ready for operation at the time.

Each piece of equipment shall be designed for the gross flow rates into and out of that equipment necessary to produce the net water flow rate from the system.

A minimum resin bed depth of 1 meter shall be used in all units. Beds deeper than 1.2 meters shall be guaranteed against attrition losses. No beds over 1.8 m deep shall be used.

When the resins become exhausted as indicated by high conductivity, or silica or based on total flow, the train shall be taken out of service and regenerated. The resins shall be backwashed, regenerated with hydrochloric acid (cation resins) and caustic (anion resins), rinsed and placed back into service or standby. Each train shall be regenerated individually. Regeneration water for

Mix Bed Polisher shall be supplied from the Raw Water storage tank.

Acid and caustic for regeneration shall be mixed with dilution water in the required proportions by adjusting metering pump stroke and dilution water flow. The required % acid or % caustic shall be monitored downstream of the mixing tees.

During low flow Mix Bed Polisher fluent will be shut down. Also the Mix Bed Polisher effluent will be recycled when the Mix Bed Polisher effluent does not meet the guaranteed performance for conductivity, pH or silica. Normally the guaranteed performance shall be met without recycling of the Mix Bed Polisher effluent.

4. DETAILED REQUIREMENTS

4.1 General

General Equipment Design Requirements:

Design and construction of the demineralized water equipment shall be suitable for location in an open building and the operating and environmental conditions under which the system will operate.

Materials of construction shall be suitable for the application and shall be consistent with the requirements of this specification.

Materials shall be identified with the applicable AWWA, ASTM, AISI, ASME etc., designation. Where no such designation is available, the manufacturer's name and the physical properties and chemical composition of the material shall be submitted for the Owner approval.

The equipment shall be shop preassembled, prepiped, prewired, shop tested, piece marked, and then broken down for shipment with consideration given for ease of erection at the Site.

Pumps shall be supplied and installed in accordance with Part 4, Specification M-006 Horizontal Centrifugal Pumps, M-007 Vertical Centrifugal Pumps, M-008 Vertical Sump Pumps and M-009 Metering Pumps.

Hoists and trolleys shall be supplied in accordance with Part 4, Specification

M-011 Miscellaneous Hoists and Trolleys.

Shop fabricated tanks shall be supplied in accordance with M-012.

4.2 Mixed Bed Polisher

The mixed bed polisher shall consist of cation exchange resin and anion exchange resin.

One cycle of regenerating shall be completed in four (4) hours approximately.

All materials selected shall be suitable for the service intended specially taking into consideration the use of concentrated and dilute hydrochloric acid and sodium hydroxide.

Mixed bed vessels shall rubber-lined carbon steel with a minimum lining thickness of 5 mm. Each vessel shall have a manhole. Access ladders shall be provided to reach the top of the tanks and these shall be welded to the tanks prior to lining the tanks.

Mixed bed vessel(s) shall have a minimum freeboard of 100% of total resin bed depth.

The outlet service header of mixed bed vessel shall be supplied with an in-line resin trap to prevent resin carry over in the event of an under drain failure. All piping to neutralization pit shall be supplied and installed.

100% spare of required amount of resins for exchange shall be included, but delivered only when requested by the Bidder at a later date.

The mixed bed polisher shall be furnished complete with perforating plate, distributor, collector, baffle plate, sight glass, manhole and all other necessary pertinents.

4.3 Make-Up Water Transfer Pumps

The pumps shall be installed outdoors at the demineralized water system area.

The pump shall be operated for the full cycle of production.

The pump shall be able to fill the water to the normal water level of make up water storage tank.

Make up water pumps and piping shall be all stainless steel construction.

The pump and motor shall be installed on the common base plate.

The motor shall be in conformity with the Electrical Technical Specification E-210 "Low Voltage Motors".

4.4 Pre-Filter

Three (3) sets of pre-filter shall be provided at the up stream of the water treatment equipment.

Two (2) set of pre-filter shall be used for normal operation and one (1) set for stand-by. Spare cartridges for 1 year of normal operation shall be furnished.

The principal material of pre-filter shall be as follows:

Filter : Paper soaked with epoxy resin, 3 micron

Shell plate : Mild steel plate, epoxy lining inside

Differential pressure indicator and switches shall be provided on each pre-filter.

4.5 Resin Carry-Over Strainers

A type 316 stainless steel wire basket strainer shall be furnished. This shall include a 5 mm rubber-lined housing with hinged and bolted cover with window, for insertion in waste discharge piping. The strainer shall be designed to retain resin fines. The Bidder shall include provisions for manually back flushing the strainer to drain.

4.6 Regeneration Equipment

The Bidder shall furnish all required regeneration equipment, consisting in general of, but not necessarily limited to the following:

4.6.1 Caustic Hot Water Tank

One (1) unit pressurized caustic dilution hot water tank of stainless steel construction, for heating and holding raw water used for resin bed preheating and for dilution of the caustic soda used for regeneration of the anion resins. The tank shall be provided with an inconel immersion heater complete with

thermostatic and loss of level controls and temperature indicator. The tank shall have installed switches for alarming abnormal temperatures - a high and low temperature alarm. A pressure regulating valve shall be provided on the tank inlet line. The tank shall be complete with relief valve and drain with immediate piping. The tank shall have an adequate capacity for consecutive regeneration of mixed bed vessels plus 25% margin, when regenerated at 45°C (120°F). The recovery time for one complete cycle in regenerating one train shall be no longer than 4 hours based on a 25°C minimum inlet temperature.

4.6.2 Acid Day Tank

FRP day tank with a bolted, sealed lid, with inlet and outlet connections with manual shutoff valves, overflow connection with a liquid seal vent connection with air desiccator, valve drain and level indicating transmitter. One high level and two low level switches shall be provided for alarm at the water treatment control panel. Filling of the day tank will be manually initiated based on the low level alarm using gravity flow or pumps. A high alarm contact shall be used to automatically trip the acid isolation valve.

The low-low level alarm contact shall be used to automatically trip the acid metering pumps. The tank shall be sized to hold the required acid for two consecutive regenerations of a mixed bed vessel, with a minimum 10% margin before reaching the low-low tank level.

4.6.3 Caustic Day Tank

A 316L SS tank, with inlet and outlet connections, manual shutoff valves, overflow connection, vent connection, valve drain and level indicating transmitter. One high level and two low level switches shall be provided for alarm at the water treatment control panel. Filling of the day tank will be manually initiated, based on low level alarm using gravity flow or pumps. The high alarm contact shall be used to automatically trip the caustic isolation valve. The low-low level alarm contact shall be used to automatically trip the caustic metering pumps. The tank shall be sized to hold the required caustic for regeneration of two consecutive regenerations of a mixed bed vessel, with a minimum 10% margin before reaching the low-low tank level.

4.6.4 Acid Metering Pumps

Two 100% capacity positive displacement diaphragm type pumps shall be

supplied in accordance with Part 4, Technical Specification, M-009 and each with sufficient capacity to regenerate one train, constructed of a high chrome nickel material ASTM B462 (UNS N08020) or ASTM A744 (GR.CN-7M) with a teflon diaphragm or equivalent.

4.6.5 Caustic Metering Pumps

Two 100% capacity positive displacement diaphragm type pumps (one spare) shall be supplied in accordance with part 4, Technical Specification M-009 and each with sufficient capacity to regenerate one train, constructed of type 316 stainless steel with a Teflon diaphragm or equivalent.

4.6.6 Mixed Bed Exchanger Regeneration Compressed Air

Oil free compressed air from Yard instrument air system shall be used for regeneration of both mixed bed ion exchangers. Pressure reducing valves and local air receiver shall be provided.

4.6.7 Notes on Procedures

Regeneration of mixed bed exchangers shall be done by a predetermined reproducible method.

4.7 Resin Design Requirements

Particle size distribution between the anion and cation resins utilized in the mixed bed vessels shall be such that no more than 10% of the cation resin beads fall within the range of anion bead sizes. Particle size distribution for the inert resin shall be equally divided between the mixed bed anion and cation bead sizes.

The relative specific gravities of the mixed bed anion and cation resins shall differ in magnitude by a minimum of 10%.

Particle size testing shall be performed in accordance with ASTM method D2187. Anion resins shall be tested in the hydroxide form.

Each resin production lot (batch) shall be shipped individually. Each lot shall be supplied with a certified analysis of the resin including, but not limited to the following parameters:

- (a) % Water Retention Capacity

- (b) % Cross linking
- (c) Wet Sieve Analysis : % retained by No. 16 sieve (1.19 mm)
% through No. 50 sieve (0.29 mm)
- (d) Exchange Capacity (kg/m³)
- (e) Friability (Grams/Bead)

The testing on sample batches shall be performed via the sample collection method as described in ASTM designation D2687-27 Method A.

The resin shall be of Rohm and Hass or Dowex quality or approved equal.

4.8 Materials

Each make-up demineralized unit shall be provided with all stainless steel internal piping, under drains, distributors etc. No gravel support beds shall be used. The material used must be suitable for hydrochloric acid and sodium hydroxide regeneration.

Chemical feed tanks required shall be designed HCl tank installed with gas seal tank. The material of storage tank shall be suitable for storage chemical. All gravity tanks shall be designed, fabricated and inspected according to API Standard 650.

Interconnecting exchanger water piping and recycle piping shall be rubber, or polypropylene lined. Dilute acid and caustic piping furnished by Bidder also shall be rubber, or polypropylene lined as well as waste piping from the exchanger vessels to the waste sump. Concentrated acid and caustic piping shall be rubber or polypropylene lined.

All exchangers shall be provided with accessible low point valve drain connections not less than 50 mm in diameter and blind flanged resin cleanout connection of 75 mm minimum Size. These nozzles shall not trap regenerant chemicals.

All air release piping and sample lines shall be stainless steel pipe or tubing.

Sufficient expansion space shall be provided above the exchange material in each mixed bed exchanger to give sample expansion during backwash.

Under drains shall be of stainless steel type as approved. Design shall be such that no supporting beds are necessary.

4.9 Demineralized Water Treatment System Piping

Demineralized water treatment system piping including interconnecting exchanger water piping, recycle piping, valves, fittings, all air release piping, other piping and piping specialties shall be in accordance with Part 4, Specification M-001 Piping, M-002 Piping Supports and Hangers, M-004 Valves and M-005 Control Valves.

Acid and caustic lines shall not be routed over aisles, walk ways and electrical equipment.

4.10 Painting, Coatings and Linings

a. The following skids shall be commercially blast cleaned in accordance with Structural Steel Painting Council (SSPC) Method SP-6 and coated with acid resistant coating :

- Acid metering pumps and day tank skid
- Caustic metering pumps and day tank skid
- All areas around the bulk storage tanks and regeneration equipment area.
- Mixed bed demineralizer skid.

b. Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and painted in accordance with manufacturer's practice.

All exposed unfinished surfaces of castings shall be properly filled, if required, before painting. All exposed machined surfaces shall be coated with a suitable anti rust compound.

All exterior surfaces of piping fittings, valves etc. other than castings, shall receive a "shop coat" of the manufacturer's standard brand of paint suitable for the service.

A band 75 mm wide shall be left unpainted in the shop along all nozzle

joints and tank sections that are to be field welded.

Field touch up painting shall be done as necessary.

After shop painting has been completed and prior to shipment, all lined vessels shall each be stenciled on the exterior side walls with a warning sign (written both in Indonesian and English) painted in bright red over white background to read as follows :

PERHATIAN - DILARANG MENGELAS ATAU MENYAMBUNG
LAS PADA TANGKI INI

LINED TANK - DO NOT WELD ON TANK OR CONNECTING
PIPING

The stencil shall be applied to at least three points around the circumference of vertical vessels and particularly on the large diameter vessels apply the stencil at points within the field of view of one another when viewed from 2 meters away.

Both inner and outer surfaces of all tank plates shall be shot-blasted to remove mill scale, and shall be painted minimum by one (1) layer of zinc-enriched primer before shipment.

After the installation of tank, welded areas and rusted parts of shop painted plates shall be re-painted by primer paint after the removal of rust scale by a grinder.

After the above work, epoxy resin base with 300 micron in thickness shall be painted on the inner surfaces, and phenolic alkyd resin base paint with 150 micron in minimum dry film thickness shall be painted in the external surface.

Painting shall be per Part 4, Specification M-001, S-420 and S-430.

5. PERFORMANCE TESTING AND GUARANTEES

The Bidder shall conduct performance testing in accordance with Part 4, Subsection 2.6 Testing and Guarantees and shall guarantee the following system guarantees :

5.1 Mixed Bed Vessels

The mixed bed demineralized vessels shall have the following guaranteed effluent water quality under all specified operating conditions:

<u>Parameter</u>	<u>Guaranteed Quality</u>
(a) Conductivity at 25 °C	< 0.1 µS/cm
(b) Silica	< 0.01 ppm
(c) Dissolved solids	< 0.01 ppm as CaCO ₃
(d) Total Fe	Less than 0.01 mg/l
(e) Total Cu	Less than 0.005 mg/l
(f) pH	6.5 - 7.5

5.2 Resin

The Bidder shall guarantee that the maximum loss of cation resin capacity, as a result of physical attrition or chemical exchangeability, shall not exceed 3% of the total initial guaranteed capacity per year for a period of 5 years.

The Bidder shall guarantee that the maximum loss of anion resin capacity including physical attrition or chemical exchangeability, shall not exceed 25% of the initial guaranteed capacity for the first twenty four months of commercial operation.

The Bidder shall guarantee the required exchange capacities and physical durability of the resins. If the specified exchange capacities cannot be maintained throughout the specified resin life, the Owner and Bidder shall agree to either to replace the complete inventory of the affected resin, or to increase the regenerant chemical dosages for the affected resin to achieve the required capacity. The Bidder shall be held liable for a percentage of the resin replacement cost equal to the percentage of the specified resin life that was not realized by the initial charge of resin, or for the additional costs of any increase in regenerant dosage over the guaranteed dosage for the remainder of the specified resin life. The amount of this liability shall be determined by the fair market of the new resins and additional chemicals at the time of their purchase.

The methods to be utilized in determining the guaranteed exchange capacities for each of the resins are described in ASTM designations D2687-77 Method Band D21 8 7-77 Methods B through J. Normally in situ regenerated resins shall be used in the total capacity determinations.

The above-mentioned exchange capacity determinations shall be performed but not the following milestones:

- (a) A sample prior to shipment but no more than one month after the resin has been manufactured.
- (b) After delivery of resin to the Site, just before placement into vessels.
- (c) After one year of operation.
- (d) After two years of operation.
- (e) After four year of operation.

All resins utilized shall be guaranteed by Bidder to withstand the raw water temperatures at the maximum desalination plant distillate temperature without degradation, or loss in capacity of the resin in any way.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance/quality control shall conform to the requirements of Part 4. Subsection 2.7.

The quality control procedures shall include, but not limited to, the following:

- Welding
- Non Destructive Testing
- Heat Treating
- Packaging and Shipping
- Continuity and High Potential Testing.

The quality control documentation shall include, but not limited to the following:

- Mill Test Reports
- NDE Test Reports
- All other documentation required by applicable codes and standards.

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall furnish and install complete instrumentation and control systems and devices for the Demineralized Water System for safe and efficient start-up and normal operation and shut down. These shall include but not limited to those listed and described in this Technical Specification.

7.2 Design Requirements

The design of instrumentation and control system and devices shall be in accordance with Part 4, Technical Specification I-001.

The design of local control panels shall be in accordance with Part 4, Technical Specification I-002.

On/off control, sequencing, interlocking and similar functions shall be carried out by a programmable logic controller PLC as described in Part 4, Technical Specification I-005. The Bidder shall provide serial link interface with plant distributed control system DCS.

The design of locally mounted instrument panels shall be in accordance with Part 4, Technical Specification I-003.

The complete operation of the equipment shall be fully monitored for automatic shutdown and alarm in case of malfunction. Failure of any operation to reach completion within a predetermined time interval shall be annunciated and shall stop the automatic process until satisfactory conditions have been reestablished.

The control system shall prevent the initiation or continuation of any automatic operation or sequence of operations which would conflict with any operation or sequence already in progress, whether such operation or sequence is under automatic or manual control.

The operating status of each make up demineralized vessel and of the external regeneration system including which automatic sequence is in progress shall be indicated by on the VDU.

Means shall be provided for manual control from the control panel of every

operation performed by the automatic control system. Such provisions shall permit manual entry into an automatic sequence, repetition under manual control of any completed operation in the sequence and return to automatic control. Controls shall be provided to permit manual extension of any regeneration step.

Signals transmitted from locally mounted transmitters to controllers, indications, recorders etc. shall be 4-20 mA DC.

The discharge to waste water system of each pump, shall be provided with pressure indication.

7.3 Functional Requirements

Instrumentation and control systems and devices shall be designed, furnished and installed to carry out safely and efficiently the functions described in Clause 3 and 4 herein including the following as a minimum.

Initiate train stop/start depending on the level in the demineralized water storage tank.

Control the outlet water temperature and inlet water pressure of the caustic hot water tank.

Control the recycling of the mixed bed effluent back to the inlet of the raw water tank during pre-starting, standby, low flow or short shut downs.

Interlock acid metering pumps with day tank level, dilution water flow and demand for regeneration.

Interlock caustic metering pumps with day tank level dilution water flow, caustic solution temperature and demand for regeneration.

Interlock the caustic dilution water with regeneration requirements and hot water tank low level.

Control the regeneration of each train in any of the following selectable modes:

Automatically based on a preset mixed bed conductivity level or a preset quantity of water treated.

Automatically after manual initiation. In this mode the controls shall automatically place the train on standby based and initiates a “Regeneration required” alarm.

Automatically carry out any regeneration step after manual initiation of that step.

Prevent simultaneous regeneration of more than one demineralized at a time and prevent automatic initiation of regeneration until the previously regenerated train is operating normally. Capability shall be included to bypass the interlock.

Alarm need for filling the acid and caustic day tanks and automatically terminate filling (after manual initiation) on high tank level. Provide level indication in the water treatment control room.

Control the start/stop of the demineralizing regeneration functions during the regeneration cycle. Interlock with water storage tank levels.

Control the pressure of the demineralizing regeneration discharge at a value suitable for the regeneration system.

Interlock the demineralized water supply pumps with and alarm low discharge header pressure. Start/stop control of these pumps will normally be carried out from the main control room. Start/stop control from the water treatment control panel shall also be provided and will be selected from the Main Control Room Remote/Local selector when operating requirements dictate.

Monitor the demineralized plant effluent conductivity. If abnormal conductivity is detected initiate an alarm, stop the flow of effluent to the storage tank and initiate the recycling function described in Clause 7.3.3. Provision shall be made for open/close control of the effluent shut off valve from the water treatment control panel. Abnormal conductivity signal shall override an open command from the control panel.

Alarm high and low level of the acid and caustic storage tanks and automatically terminate filling on high tank level.

Control the start/stop of the water treatment supply pumps to suit the demand for demineralized water and interlock with the raw water storage tank level.

7.4 Details

7.4.1 Sampling

Stainless steel sample connections shall be provided for sampling the influent and effluent of each ion exchange vessel, and the dilute chemical regeneration lines.

These samples shall contain an isolation valve at the source and shall terminate with a valve at the sample panel specified in Part 4, Technical Specification M-406. The following samples shall be continuously monitored for specific conductance when in operation.

- Mixed Bed Effluent
- Dilute Acid Regenerant
- Dilute Caustic Regenerant

In addition, the mixed bed effluent shall be continuously monitored for pH, conductivity and silica.

7.4.2 Analyzers

All analyzers shall be provided with mechanical pre-filters.

Analyzers shall provide 4-20 mA DC signals for control, indication and recording purposes.

Analyzers for pH shall have a stainless steel flow through chamber with glass measuring electrodes, temperature compensation and gel filled reference electrodes. They shall be Leeds and Northrup Model 7773 or approved equal. Ultrasonic cleaning shall be provided for the sensor where required.

Analyzers for specific conductance shall have Leeds and Northrup Model 4905 (or approved equal) flow through cells with temperature compensation and platinum electrodes bell constants shall be suitable for the ranges of the application. Cells shall be provided with cell holders complete with terminal blocks and doors or removable cover plates. Conductivity monitors shall be Leeds and Northrup Model 7073 (or approved equal) with indicator ranges suitable for the application.

Silica analyzers shall be by Scientific Instruments Inc of New York or

Technician or approved equal.

7.4.3 Local Control Panels

The Bidder shall furnish and Install local control panels adjacent to each piece or group of equipment as required. The local control panels shall contain all instruments and devices necessary for local control and test purposes for the associated equipment.

Refer to Part 4, Technical Specification I-002 and I-003 for Control panel for centralized coordinated control.

7.4.4 Instrumentation and Control Devices

The Bidder shall furnish and install all locally mounted instrumentation and control devices necessary for the safe and efficient operation of the system including and required to perform the functions listed in Clause 7.3. Non-metallic instrument air tubing shall not be allowed. All tubing shall be supported continuously to the maximum practicable extent to prevent physical damage.

8. ELECTRICAL SPECIFICATION

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Demineralized Water System.

Those equipment shall include but not be limited to the followings :

- (a) Motors
- (b) Power panel
- (c) Conduits, cables and wiring.
- (d) All electrical control, monitoring and instrumentation devices, and other appurtenances required for this specification, M-401 shall be provided by the Bidder.

For general electrical requirements, see Part 4, Technical Specification E-001.

8.2 Motors

For detailed electrical requirements, see Part 4, Technical Specification E-210 for Low Voltage Motors.

8.3 Power Panel

For detailed electrical requirements, see Part 4, Technical Specification E-001 and E-280.

The panel shall have a degree of protection IPW 55 of IEC 60529.

8.4 Conduits, Cables and Wiring

All motors and electrical, control, monitoring and instrumentation devices furnished and installed by the Bidder shall be wired by him to the power and local control panel respectively.

The wiring between those panels shall also be included.

The required power, control, instrumentation and thermocouple extension cables shall be provided by the Bidder.

For detailed electrical requirements, see Part 4, Technical Specifications E-230 E-231 and E-232.

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